

1. Users Collection

Stores information about system users.

```
{
  "_id": "ObjectId",          // MongoDB unique ID
  "name": "John Doe",         // Full name
  "email": "john@example.com", // Login email
  "password_hash": "hashed_password", // Hashed password for authentication
  "organization_id": "org_123", // Links user to an organization
  "role": "student",          // student / employee / staff / admin
  "status": "active",         // active / inactive / suspended
  "last_login": "2026-03-29T09:30:00Z",
  "created_at": "2026-03-29T08:00:00Z",
  "updated_at": "2026-03-29T09:00:00Z"
}
```

Notes:

- Passwords should never be stored in plaintext.
- last_login and status help in monitoring active users.

2. Organizations Collection

Stores organization-level metadata.

```
{
  "_id": "org_123",
  "name": "COMSATS University",
  "domain": "comsats.edu.pk",
  "admin_users": ["user_001", "user_002"], // IDs of admin users
  "created_at": "2026-03-01T10:00:00Z",
}
```

```
"updated_at": "2026-03-10T15:00:00Z",  
"settings": {  
  "language": "en",  
  "max_document_size_mb": 50,  
  "allow_document_upload": true  
}  
}
```

Notes:

- settings allow customization per organization.
- admin_users help with managing document uploads or approvals.

3. Documents Collection

Stores uploaded document metadata.

```
{  
  "_id": "doc_456",  
  "organization_id": "org_123",  
  "title": "Academic Regulations 2024",  
  "description": "Regulations for attendance and exams",  
  "file_name": "academic_regulations.pdf",  
  "file_type": "pdf",           // pdf, docx, txt  
  "file_size_mb": 2.5,  
  "upload_user_id": "user_001", // Who uploaded  
  "upload_date": "2026-03-29T10:05:00Z",  
  "processed": true,           // Has it been chunked & embedded  
  "chunk_count": 12,           // Number of chunks created  
  "tags": ["academic", "policy"], // Optional tags for analytics  
}
```

```
"version": 1,           // Version control for updated docs
"last_updated": "2026-03-29T10:10:00Z"
}
```

Notes:

- processed flag helps to track documents ready for FAISS embedding.
 - chunk_count is useful for QA and analytics.
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4. Document Chunks Collection

Stores segmented document text and links to FAISS embeddings.

```
{
  "_id": "chunk_789",
  "document_id": "doc_456",
  "organization_id": "org_123",
  "chunk_index": 1,           // Sequence in document
  "text_chunk": "Students must maintain 75% attendance...",
  "embedding_id": "faiss_001", // FAISS reference
  "embedding_model": "sentence-transformers/all-MiniLM-L6-v2",
  "tokens_count": 25,         // Number of tokens in this chunk
  "created_at": "2026-03-29T10:06:00Z",
  "last_updated": "2026-03-29T10:06:00Z"
}
```

Notes:

- chunk_index ensures correct ordering if needed.
 - embedding_id links chunk to FAISS vector for retrieval.
 - Storing tokens_count helps control LLM context limits.
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5. Queries Collection

Stores each user query and results.

```
{
  "_id": "query_001",
  "user_id": "user_123",
  "organization_id": "org_123",
  "query_text": "When is the semester fee deadline?",
  "predicted_category": "finance",
  "retrieved_chunks": [
    "chunk_789",
    "chunk_790"
  ],
  "retrieved_documents": [
    "doc_456"
  ],
  "generated_answer": "The semester fee deadline is 31st May 2026.",
  "sources": [
    "Academic Regulations 2024, Section 3.2"
  ],
  "feedback": null,          // User can rate helpfulness
  "response_time_ms": 250,   // Response latency
  "timestamp": "2026-03-29T10:10:00Z",
  "status": "answered"      // answered / unresolved
}
```

Notes:

- retrieved_chunks store IDs to trace the source of LLM responses.

- feedback allows future improvements of answers.
 - response_time_ms helps monitor system performance.
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6. Optional Analytics / Logging Collection

For usage monitoring and ML insights.

```
{  
  "_id": "log_001",  
  "organization_id": "org_123",  
  "user_id": "user_123",  
  "action": "query_submitted",    // Could also be document_upload, login  
  "details": {  
    "query_length": 7,  
    "category": "finance"  
  },  
  "timestamp": "2026-03-29T10:10:00Z"  
}
```

Notes:

- This helps track frequent queries, user engagement, and ML performance.